# AI-Enabled Smart Ambulance System for Rapid Emergency Response Using IoT

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***Abstract*—AI-enabled Smart Ambulance System is designed to improve emergency health care services by reducing response time and improving coordination.In day to day life delays often occur due to traffic congestion,poor communication and lack of real time patient information.In these critical condition reaching to the patient is very much important.Hence our proposed system which uses IOT based medical sensors to capture health parameters such as heart rate,body temperature and oxygen levels and sends these real time data to hospitals.The system also uses GPS technology to track ambulance locations in order to get accurate arrival time . In addition to this the system helps to decide best optimal route for ambulance to reach the destination quickly.This ensures smooth movement even in crowded areas.As told earlier poor communication is one of the problem in delayed process,our architecture supports better communication between ambulances and hospitals by sharing real time updates like heartbeat rate , body temperatures etc.By this way doctors can prepare of treatment before patient's arrival.This reduces waiting time and improves quality of emergency care. Furthermore, the system can assist in making quick decisions during emergencies by analyzing incoming data and suggesting appropriate actions. This makes the overall response more efficient and reliable. Overall, the proposed approach improves emergency response efficiency, enhances coordination, and has the potential to save more lives by providing faster and smarter medical assistance.**

***Index Terms*—– AI-enabled decision support, intelligent ambulance routing, IoT medical sensors, traffic signal control, emergency healthcare**

## I. Introduction

Emergency Medical Services (EMS) are very important in saving lives during serious situations such as road accidents, heart attacks, and other life threatening conditions.However these Medical services may not work as efficiently as they should.This is mainly due to heavy traffic, poor route planning, and lack of proper coordination between ambulances, traffic systems and hospitals.Because of these issues ambulances take more time to reach the incidents and transport them back to hospitals.This may lead to Severe problems and also human loss may occur. One important concept in emergency care is the golden hour, which refers to the first hour after a serious injury.If proper treatment is given within this period ,the chances of survival is more.But our traditional ambulance system depends on manual decisions and fixed traffic signals, which cannot adjust quickly to real time conditions. To overcome these challenges, this paper proposes an AI-enabled smart ambulance system that uses modern technologies such as Artificial Intelligence (AI), Internet of Things (IoT) and real time communication. The main goal of the system is to improve emergency response by increasing speed, accuracy and coordination between ambulances, traffic control systems and hospitals.One of the main features of the system is intelligent traffic management.The system uses AI algorithms to study real time traffic conditions and select the fastest and optimal route for the ambulance. The system also controls the traffic signals by giving first priority to ambulances to move quickly without having unnecessary stops.As told earlier another important feature is real time patient monitoring.The system is equipped with sensors that detect patients conditions and convert that data into documentations to send it to hospitals in real time.So that doctors can understand patients conditions before they reach to hospitals.In addition, the system includes a hospital alert feature that informs nearby hospitals about the emergency, along with patient details and estimated arrival time. This helps hospitals stay ready and reduces delay in treatment.Overall, this system provides a more connected and efficient approach to emergency healthcare. It helps reduce response time, improves patient care and supports the development of smarter and more reliable healthcare systems.

## II. Literature Survey

The literature survey provides a detailed understanding of various technologies used in smart ambulance systems, intelligent traffic management and healthcare monitoring solutions. It highlights the progress made in this domain and identifies the limitations of existing approaches. Most of the current systems focus on solving only a single aspect of the problem, rather than providing a fully integrated solution that combines routing, monitoring, communication, and traffic control.

J. Naarayanan et al. [1] proposed an Artificial intelligence based smart ambulance system with patient monitoring and decision support features. Their system has improved decision making and signal control.But it lacked effective route optimization and real time traffic signal coordination. Similarly, H. Nozari et al. [2] developed a routing optimization model using Machine Learning and Genetic Algorithms, which helped reduce ambulance response time. But, their work was limited to simulation and did not consider real time traffic signal interactions.

S. Verma et al. [3] introduced an IoT and cloud based ambulance system that enabled remote monitoring of patients. While their system improved patient data accessibility, it lacked intelligent routing capabilities. R. Gupta et al. [4] focused on real time ambulance tracking, which had improved location accuracy, but their system did not include route optimization or hospital alert mechanisms. P. K. Sharma et al. [5] worked on a blockchain-based communication framework that enhanced data security in healthcare systems, but routing and traffic management aspects were not addressed.

S. B. Prakash et al. [6] developed a system for traffic signal prioritization, allowing ambulances to move quickly through intersections. However, patient monitoring and route planning were not included in their approach. N. Kumar et al. [7] proposed a VANET-based system to reduce waiting time at traffic signals, but it heavily depended on network availability and lacked integration with healthcare systems.

A. K. Mishra et al. [8] worked on traffic congestion prediction using machine learning techniques, but their model was not specifically designed for emergency vehicles. S. R. Bhoi et al. [9] created a priority based traffic control system for emergency vehicles. But, it lacked coordination with hospital systems. M. A. Jabbar et al. [10] developed a data mining based ambulance dispatch system, which improved decision making, but did not fully consider real time traffic conditions.

A. K. Sahu et al. [11] focused on cloud based healthcare monitoring, enabling real time data sharing between devices and medical systems, but routing was not included. R. Singh et al. [12] applied deep learning techniques for traffic density estimation, but their work did not specifically target ambulance systems. V. R. Balaji et al. [13] used Genetic Algorithms for route optimization however, their system lacked traffic signal control mechanisms.

K. Lakshmi et al. [14] developed an AI based ambulance system with hospital alert mechanisms, which improved pre notification to hospitals, but had limited traffic intelligence. P. Rajput et al. [15] worked on Internet of Things based patient monitoring systems, but their approach did not include routing or coordination features. S. K. Panda et al. [16] proposed adaptive traffic signal control using machine learning, but it was not specifically designed for emergency vehicle prioritization.

A. N. Jha et al. [17] developed an edge computing based healthcare system that reduced latency in data processing, but it lacked traffic optimization features. R. K. Singh et al. [18] introduced a 5G based communication framework for ambulances, which improved connectivity and communication speed, but did not include routing mechanisms. P. S. Rathore et al. [19] proposed an AI based routing model that improved route accuracy, but did not integrate hospital coordination. Finally, V. Patel et al. [20] developed a dispatch optimization system that reduced travel time, but lacked patient monitoring and real time communication features.

From these studies, it is evident that most existing systems focus on individual components such as routing, monitoring, communication, or traffic signal control. Very few approaches combine all these features into a single integrated system. This creates a clear gap in current research. Therefore, there is a need for a comprehensive solution that integrates AI based routing, real time traffic signal control, continuous patient monitoring and hospital coordination. Such a system can significantly improve emergency response efficiency and contribute to saving more lives.

## III. Problem Formulation

### A. Problem Description

Emergency medical services are essential for saving lives during serious situations such as road accidents, heart attacks, strokes and other critical conditions.Our current ambulance systems often face delays due to heavy traffic and poor route planning.As cities grow traffic congestion increases and fixed traffic signals make it difficult for ambulances to move quickly.In many cases ambulance drivers depend on manual decisions to choose routes without access to real time traffic information.At the same time hospitals usually receive only limited details about the patient which delays preparation for treatment.Another major issue is the lack of coordination between ambulance systems, traffic control systems and hospitals.Because these systems work separately, the overall emergency response becomes slow and inefficient.To overcome these challenges there is a need for an intelligent system that combines AI based routing, smart traffic signal control and real time communication.Such a system can reduce delays, improve coordination and increase the chances of patient survival, especially during the critical golden hour.

### B. Problem Statement

The current ambulance system suffers from traffic congestion, lack of intelligent routing and poor coordination with traffic signals and hospitals.This leads to delays in emergency response and reduces the chances of timely medical care.Additionally there is no proper real time monitoring of patient conditions during transit.Therefore there is a need for an integrated smart system that can improve routing, enable real time monitoring and enhance coordination .This provides faster and more effective emergency services.

### C. Objectives

- AI Based Smart Ambulance System: To develop a system that uses artificial intelligence to improve ambulance decision making and provide faster and more efficient emergency services.

- Smart Traffic Signal Integration: To design a system that controls traffic signals intelligently and selects the best route which allows ambulances to move quickly with minimal delays.
- Improved Emergency Response: To reduce response time, improve patient care and increase survival chances by ensuring better coordination between ambulances, traffic systems and hospitals.

## IV. Proposed Methodology

### A. System Architecture


Fig. 1. System Architecture of Smart Ambulance System

Our proposed smart ambulance system is designed using an integrated and multi layered architecture that connects emergency services, traffic systems and healthcare units. Our system includes key components such as the emergency caller interface, control room, ambulance system, traffic signal system, cloud server and hospital module.The architecture ensures continuous communication between all components. When an emergency request is received, the control unit processes the request and assigns the nearest available ambulance. The ambulance system then activates AI based route optimization to determine the fastest path using real time traffic data.The system also communicates with traffic signals to provide priority access, creating a smooth path for ambulance movement.At the same time IoT sensors inside the ambulance collect patient health data and send it to the cloud server. The cloud server stores and forwards this data to hospitals enabling early medical preparation.This design ensures better coordination, reduced delays and improved emergency response efficiency.

### B. Working Procedure

The working procedure of the proposed Smart Ambulance System follows a sequence of coordinated steps designed to reduce response time and improve patient care during emergencies.

The process begins when an emergency call is received from a user or a medical support system. This request is immediately captured by the control system, where the details such as location, type of emergency, and urgency level are analyzed. Based on this information, the system identifies the most suitable ambulance available in the nearby region.

Once the nearest ambulance is selected, it is dispatched without delay. At the same time, the system activates an AI-based route optimization mechanism, which calculates the fastest and most efficient path by considering real-time traffic conditions. This helps the ambulance avoid congested routes and reach the destination quickly.

To further improve movement, the traffic signal control system is triggered to create a green corridor. Traffic signals along the selected route are automatically adjusted, allowing the ambulance to pass through intersections without stopping. This significantly reduces travel delays in urban traffic conditions.

During transit, the patient's vital parameters such as heart rate, body temperature, and oxygen levels are continuously monitored using IoT-based medical sensors installed in the ambulance. These sensors collect real-time data and ensure that the patient's condition is tracked throughout the journey.

The collected data is transmitted to a cloud server, which acts as a central platform for storing and sharing information. From the cloud, the data is forwarded to the hospital, where medical staff can access the patient's condition in advance. The system also provides an estimated time of arrival (ETA), helping the hospital prepare accordingly.

With this prior information, doctors and medical staff can arrange the necessary equipment and plan the required treatment before the patient arrives. This preparation plays a crucial role in reducing critical treatment delays.

Finally, the ambulance reaches the hospital, and the patient is immediately attended to with minimal waiting time. The overall process ensures faster response, better coordination, and improved emergency healthcare delivery.

### C. Modules

The proposed system is divided into multiple functional modules, each responsible for handling a specific part of the emergency response process.

**Emergency Management Module:** This module is responsible for receiving and processing emergency requests. It captures user inputs, identifies the nature of the emergency, and initiates the response workflow.

**Ambulance Management Module:** This module maintains information about available ambulances, including their current location and status. It helps in selecting and dispatching the nearest ambulance efficiently.

**Route Optimization Module:** This module uses Artificial Intelligence techniques to determine the best possible route for the ambulance. It considers real-time traffic data and continuously updates the route if conditions change.

**Traffic Signal Control Module:** This module manages traffic signals to create a green corridor for the ambulance. It ensures smooth movement by reducing stops at intersections.

**Patient Monitoring Module:** This module collects real-time health data using IoT sensors. It monitors vital parameters and ensures continuous tracking of the patient's condition during transit.

**Cloud Communication Module:** This module handles data transmission between the ambulance, cloud server, and hospital. It ensures secure and real-time data exchange.

**Hospital Alert Module:** This module sends patient details and estimated arrival time to the hospital. It allows medical staff to prepare in advance for treatment.

**Admin/System Management Module:** This module is responsible for overall system control and monitoring. It manages user access, system updates, and performance tracking.

### D. Implementation Details

The proposed system is implemented using a combination of modern technologies to ensure efficient and real time performance. The user interface for emergency requests and monitoring is developed using React.js, which provides a responsive and user friendly experience. The backend system is implemented using FastAPI (Python), which handles request processing, ambulance dispatch and communication between system components through REST APIs. For traffic analysis, our system uses YOLO based computer vision, which detects vehicles and analyzes traffic density in real time. This helps in improving route optimization and traffic signal control. The routing module integrates GPS with AI algorithms to determine the fastest and least congested path for ambulances. The patient monitoring system uses ESP32 microcontroller with sensors to collect real time health data such as heart rate, temperature and oxygen levels. Data communication between ambulance and hospital is handled through cloud based servers using HTTP protocols, ensuring fast and secure data transfer.The hospital system is implemented as a web based dashboard, which receives patient data and alerts allowing medical staff to prepare treatment in advance. These technologies together ensure that the system is scalable, efficient and capable of handling real time emergency situations.

TABLE I
SUMMARY OF METHODOLOGY COMPONENTS

| | |
|---|---|
| **Component** | Emergency Interface |
| **Technology** | React.js |
| **Purpose** | User interaction and emergency request submission |
| **Standards** | WCAG 2.1 AA |
| **Component** | Control System |
| **Technology** | FastAPI (Python) |
| **Purpose** | Handles requests and ambulance dispatch |
| **Standards** | REST API standards |
| **Component** | Traffic Management |
| **Technology** | YOLO + Computer Vision |
| **Purpose** | Detects vehicles and analyzes traffic density |
| **Standards** | Not Applicable |
| **Component** | Routing Module |
| **Technology** | GPS + AI Algorithms |
| **Purpose** | Finds fastest and least congested route |
| **Standards** | Not Applicable |
| **Component** | Patient Monitoring |
| **Technology** | ESP32 + Sensors |
| **Purpose** | Collects real-time patient vital data |
| **Standards** | IoT Standards |
| **Component** | Cloud Communication |
| **Technology** | HTTP + Cloud Server |
| **Purpose** | Transfers data between ambulance and hospital |
| **Standards** | Secure Communication |
| **Component** | Hospital System |
| **Technology** | Web Dashboard |
| **Purpose** | Receives patient data and alerts |
| **Standards** | Healthcare Interoperability |

## V. Results and Discussion

### A. Overview

The proposed Smart Ambulance System improves emergency response by utilizing Artificial Intelligence, Internet of Things and real time communication technologies to enhance overall efficiency. One of the key outcomes of the system is the significant reduction in ambulance response time, enabling faster arrival at emergency locations. This improvement is achieved through intelligent route selection and smart traffic signal control, which help ambulances avoid traffic congestion and reach their destination more quickly. As a result, the system ensures timely medical assistance and plays an important role in improving the effectiveness of emergency healthcare services.

Our system results in enhancement of patient care during transit. The system uses IoT based devices to continuously monitor vital signs such as heart rate, temperature and oxygen levels. It sends this data to the hospital in real time, allowing medical staff to assess the patient's condition before arrival. This helps doctors prepare the required treatment in advance and ensures a faster medical response. As a result, delays in critical care are reduced and patient outcomes are improved.

### B. Performance Analysis

TABLE II
RESULTS AND DISCUSSION SUMMARY

| Aspect | Results | Discussion |
|---|---|---|
| Response Time | Reduced ambulance arrival time | Achieved using AI routing and signal priority |
| Route Optimization | Faster route selection | Real-time data improves decisions |
| Patient Monitoring | Continuous vital tracking | Helps hospitals prepare early |
| Communication | Better coordination | Real time sharing reduces delays |
| Hospital Preparedness | Improved readiness | Reduces treatment delay |
| System Efficiency | Overall performance improved | AI, IoT, cloud integration |
| Limitations | Depends on network | Needs proper infrastructure |
| Scalability | Suitable for smart cities | Can extend with 5G |

The system improves coordination between ambulances, traffic control and hospitals. It enables real time communication, allowing all components to share information quickly and make better decisions. This approach reduces delays, minimizes errors and improves overall efficiency.

### C. Graphical Analysis

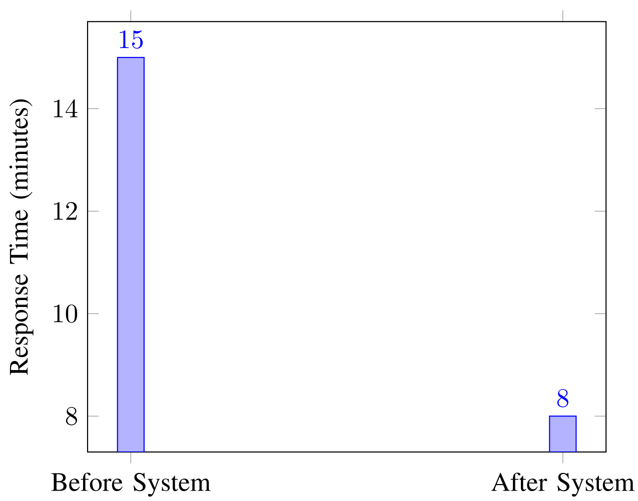


Fig. 2. Comparison of ambulance response time before and after system implementation

The graph clearly shows a reduction in ambulance response time after implementing the proposed system. This improvement is mainly due to intelligent routing and traffic signal prioritization.

### D. Experimental Results

The implementation of the proposed system is validated through experimental snapshots that demonstrate its functionality and performance. Fig. 3 illustrates the frontend interface, where users can upload traffic images and initiate the detection process. Fig. 4 presents the output of the YOLO-based vehicle detection model, in which vehicles are accurately identified and highlighted using bounding boxes. Fig. 5 shows the backend API response, displaying the detected vehicle count and processed data.

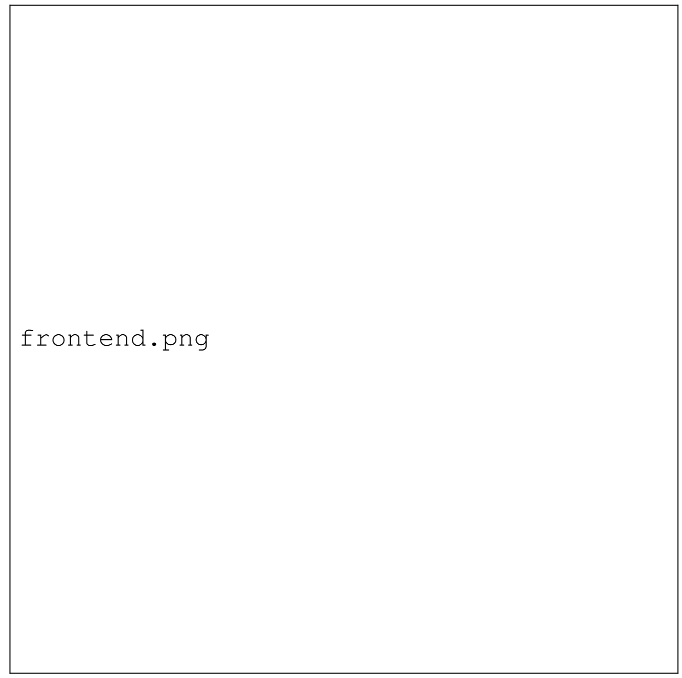


Fig. 3. Frontend interface used for uploading traffic images and initiating vehicle detection

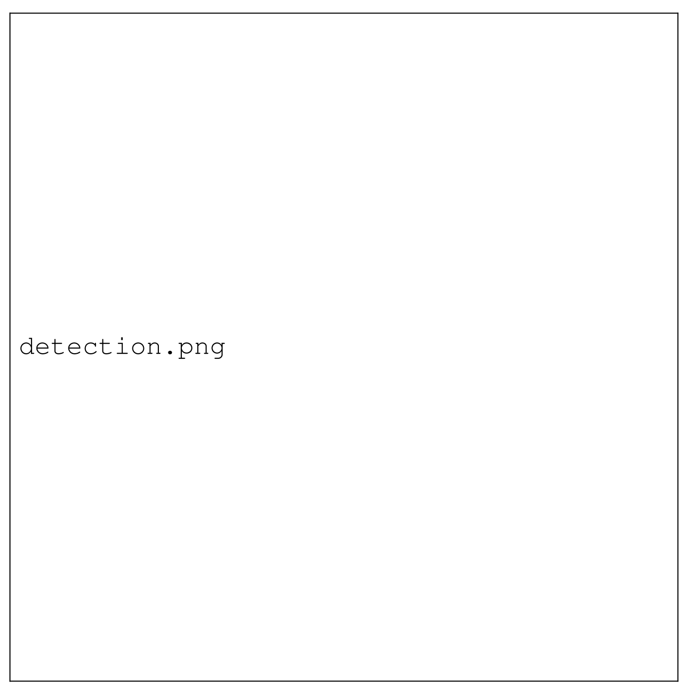


Fig. 4. Vehicle detection output using YOLO model showing identified vehicles and bounding boxes

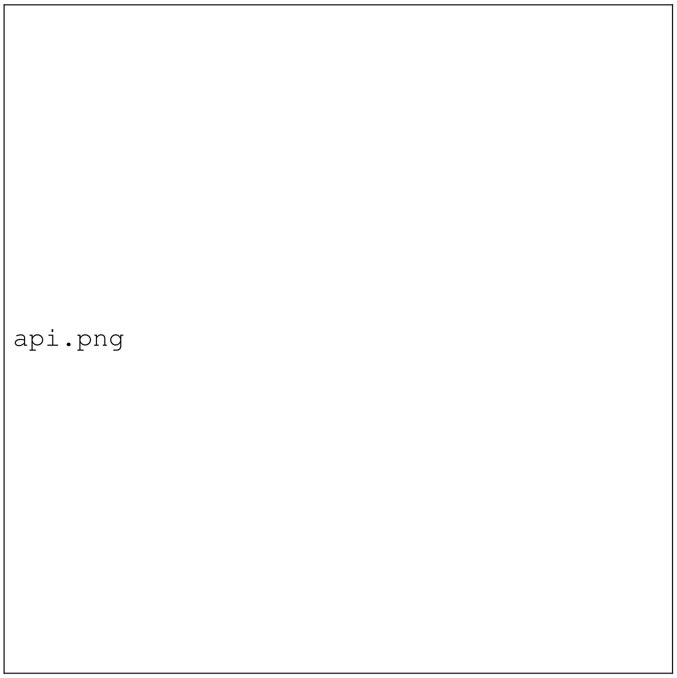


Fig. 5. Backend API response displaying detected vehicle count and processed data

### E. Discussion

From a our perspective, the system demonstrates strong potential for deployment in modern smart city environments. The system depends on reliable network connectivity, proper infrastructure and smooth integration with existing traffic management systems.With these challenges, our system provides a scalable and efficient solution that improves emergency healthcare services and increases patient survival rates.

These results confirm that our system is capable of performing real time traffic analysis efficiently. The integration of computer vision techniques with backend processing enables accurate vehicle detection and quick data handling. This ensures that traffic conditions are analyzed in a timely manner, supporting faster decision making. The experimental outcomes demonstrate the system's effectiveness in improving traffic management and enhancing emergency response operations.

## VI. Conclusion

The proposed Smart Ambulance System addresses one of the most critical challenges in emergency healthcare which is delay in ambulance response. By combining Artificial Intelligence, IoT and real time communication, the system creates a coordinated environment where ambulances, traffic signals and hospitals work together efficiently.

The system enables ambulances to move faster by selecting optimal routes and prioritizing traffic signals, reducing unnecessary delays. At the same time it continuously monitors patient health using IoT based sensors and sends this data directly to the hospital. This allows medical staff to understand the patient's condition in advance and prepare treatment before arrival.

The integration of these components improves coordination between all parts of the emergency response system. It reduces response time, minimizes errors and ensures that patients receive timely care. The results show that the system not only improves operational efficiency but also has the potential to increase patient survival rates.

The system depends on reliable network connectivity and proper infrastructure also it provides a strong foundation for smart city healthcare solutions. With further enhancements and real world deployment, the system can play a significant role in transforming emergency medical services.

## Future Work

The proposed system can be further improved by introducing advanced technologies that enhance both its performance and reliability in real world scenarios. One important direction is the integration of 5G communication, which can support faster and more stable data exchange between ambulances, traffic systems and hospitals. With improved connectivity, critical information such as patient data and route updates can be shared without delay, which is essential during emergency situations.

Another area of improvement is the use of edge computing. Instead of sending all data to a central server, processing can be done closer to the source, such as within the ambulance itself. This helps in reducing latency and allows quicker responses when immediate decisions are required. It also reduces dependency on network conditions, making the system more robust.

The system can also be enhanced by incorporating wearable health monitoring devices. These devices can continuously track vital signs such as heart rate, oxygen levels and temperature during transit. By providing more accurate and real time data they help medical staff understand the patient's condition better before arrival.

In addition to this, predictive analytics and machine learning techniques can be applied to improve decision making. By analyzing past traffic patterns and emergency data, the system can anticipate congestion and possible delays. This allows the system to choose better routes and prepare for different scenarios in advance improving overall efficiency.

For large scale deployment the system can be integrated with existing smart city infrastructure. By connecting with urban traffic systems and centralized control units it becomes easier to manage emergency vehicles across different regions. Such integration will make the system more practical for real world implementation.

These enhancements can make the system more flexible reliable and capable of handling complex emergency situations. With continuous development and proper deployment the system has the potential to significantly improve emergency healthcare services and response efficiency.

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